

A Federated Learning Approach for Layer 2 Network Intrusion Detection with Explainable AI Across Distributed Network Segments

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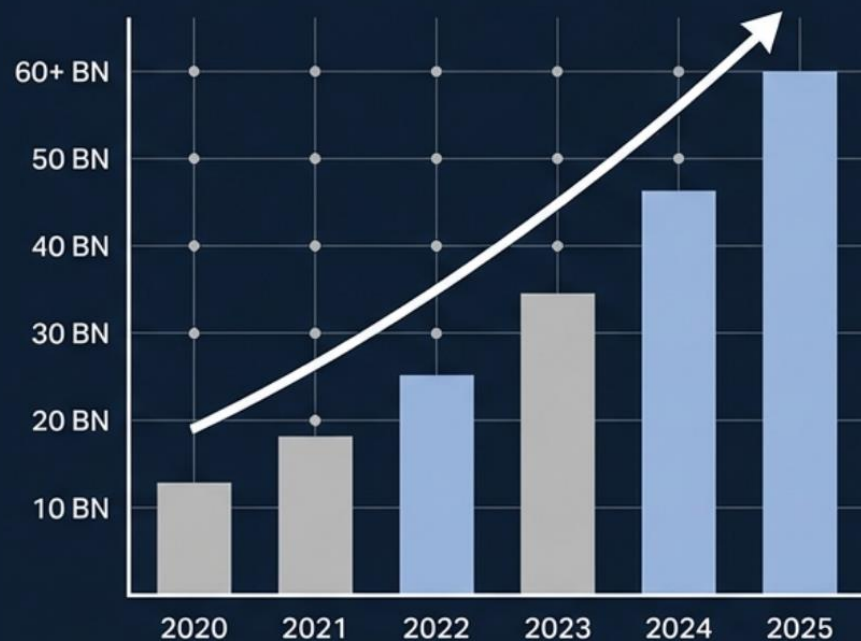
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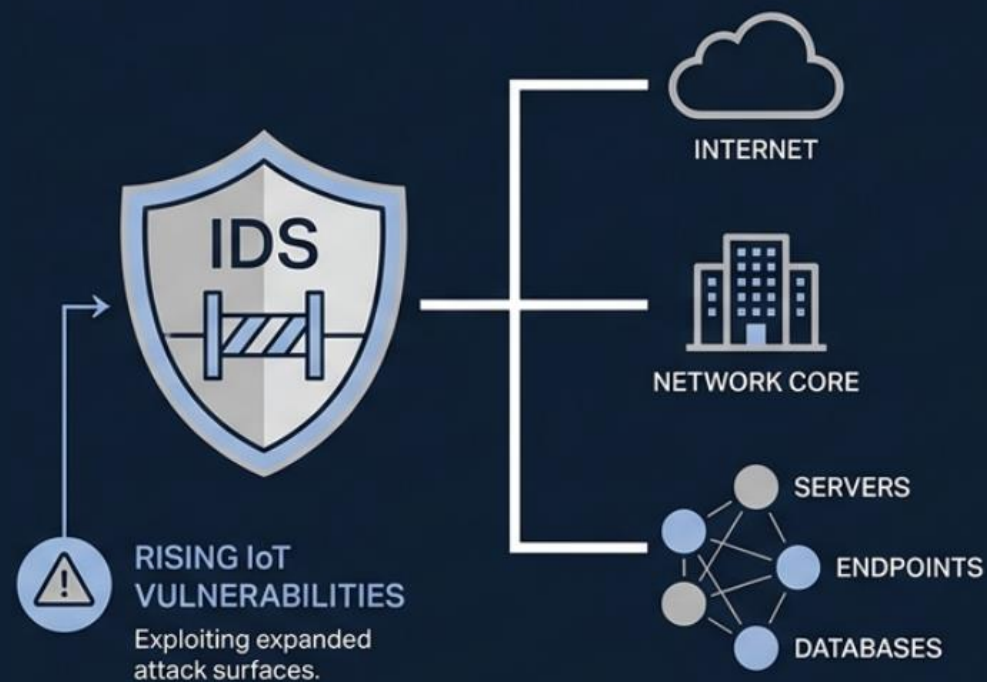
THE THREAT LANDSCAPE

Exponential growth of edge devices



Connected devices surging globally.

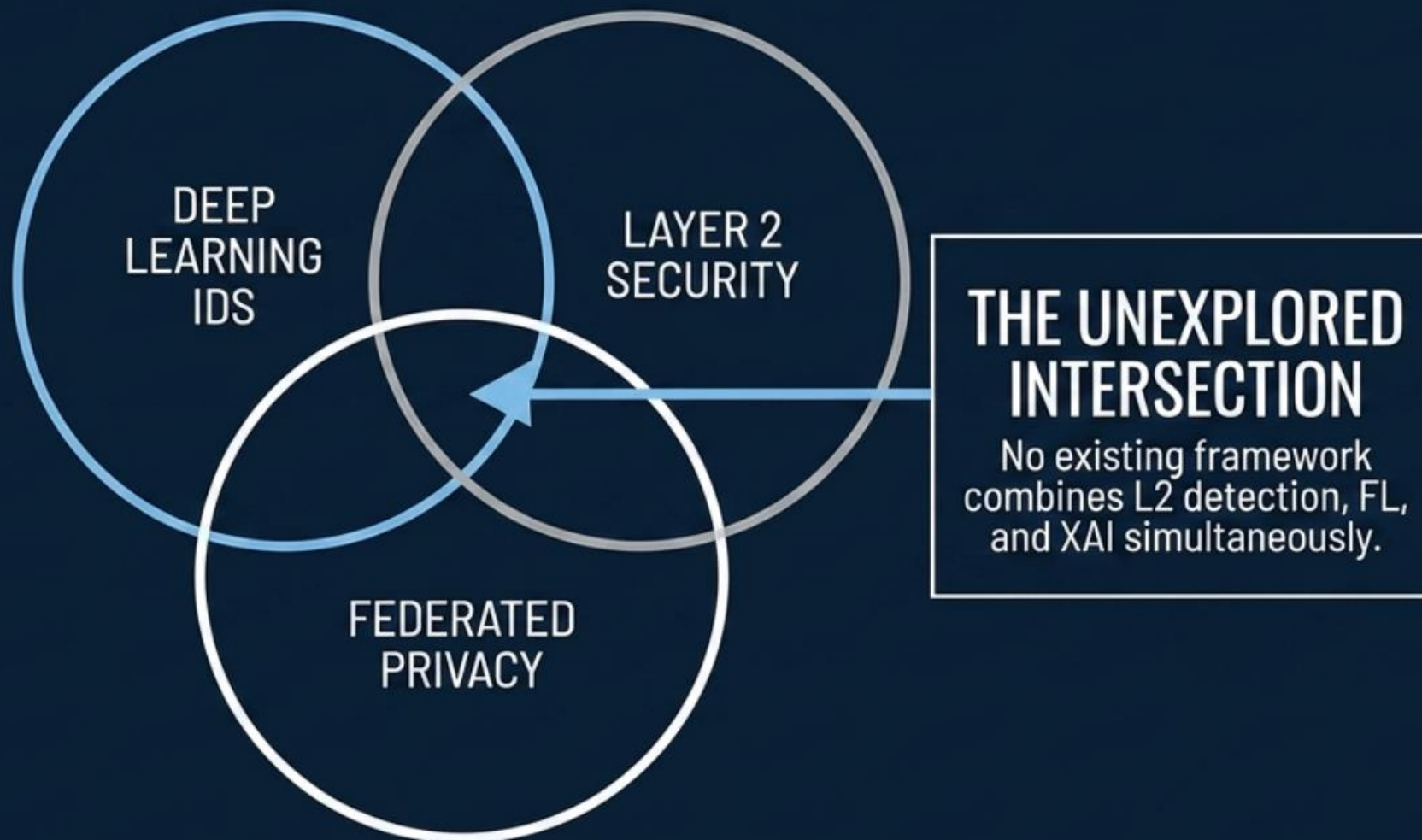
TRADITIONAL IDS AS THE PERIMETER GATEKEEPER



THE LAYER 2 BLIND SPOT



IDENTIFYING THE RESEARCH GAP



SIGNIFICANCE OF THE STUDY



Security

Halts widespread lateral damage by detecting hidden L2 threats early.



Privacy

Eliminates the transmission of sensitive network packets to centralized servers.



Scientific Contribution

Pioneers a foundational framework combining Edge AI, FL, and XAI.

RESEARCH OBJECTIVES

1

Maximize Detection Accuracy

Detect localized Layer 2 attacks with high precision.

3

Preserve Data Privacy

Keep raw data on local nodes at all times.

2

Ensure Explainability

Provide transparent, human readable reasoning for alerts.

4

Lightweight Deployment

Operate efficiently on edge hardware with limited resources.

CORE RESEARCH QUESTIONS

Q1

Can a federated model match the accuracy of a centralized model?

Q2

How does Non-IID data distribution impact model performance?

Q3

Is SHAP effective at identifying key L2 attack features?

Q4

Can this system execute on lightweight hardware like Raspberry Pi?

RESEARCH HYPOTHESES

H1

CNN + Bi-LSTM Architecture

Will maximize the accuracy of stealthy attack detection

H2

FedProx over FedAvg

Will stabilize training and perform better on Non-IID distributed data.

H3

SHAP Integration

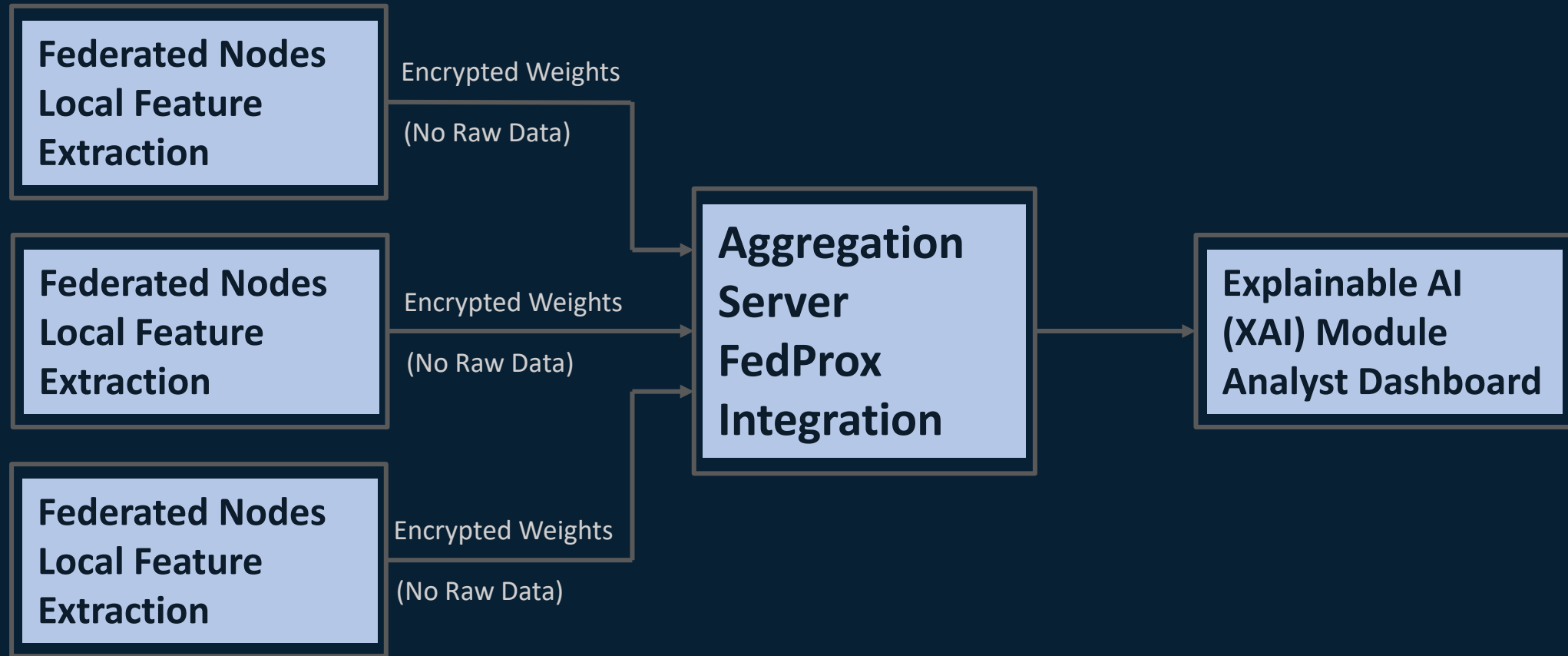
Will successfully demystify the AI black box
And increase decision transparency.

LITERATURE MATRIX

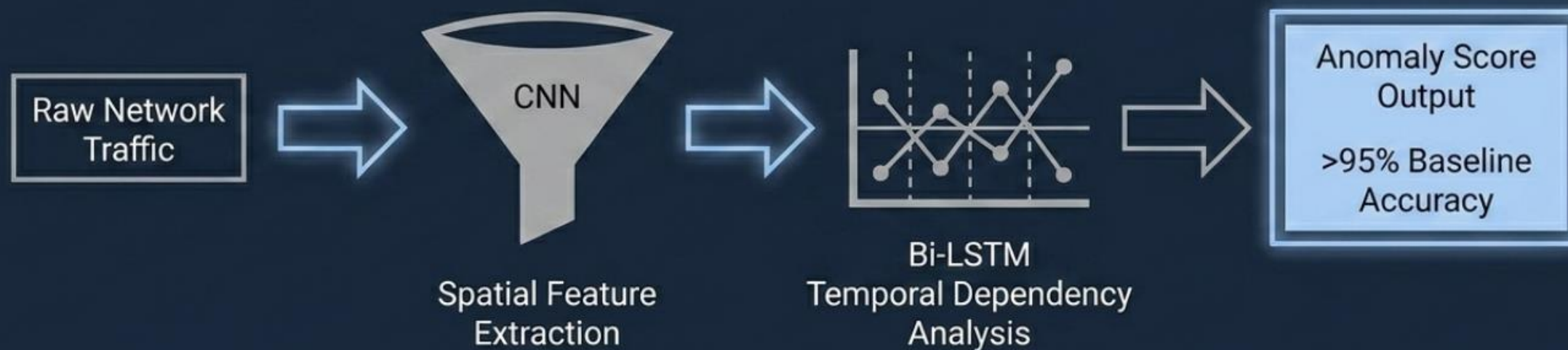
Study	L2 Detection	Deep Learning	Federated Privacy	Explainable AI
Papers 1-2	✓	✓	✗	✗
Papers 3-4	✓	✓	✗	✓
Papers 5-7	✗	✗	✓	✗
Proposed Framework	✓	✓	✓	✓

Review of 7 foundational papers confirms the absence of a comprehensive IDS solution.

PROPOSED SYSTEM ARCHITECTURE



THE DEEP LEARNING ENGINE



FEDERATED LEARNING FRAMEWORK

HANDLEING REAL-WORLD NON-IID DATA

FEDAVG ALGORITHM

UNSTABLE CONVERGENCE



Struggles with uneven data distribution across network segments.

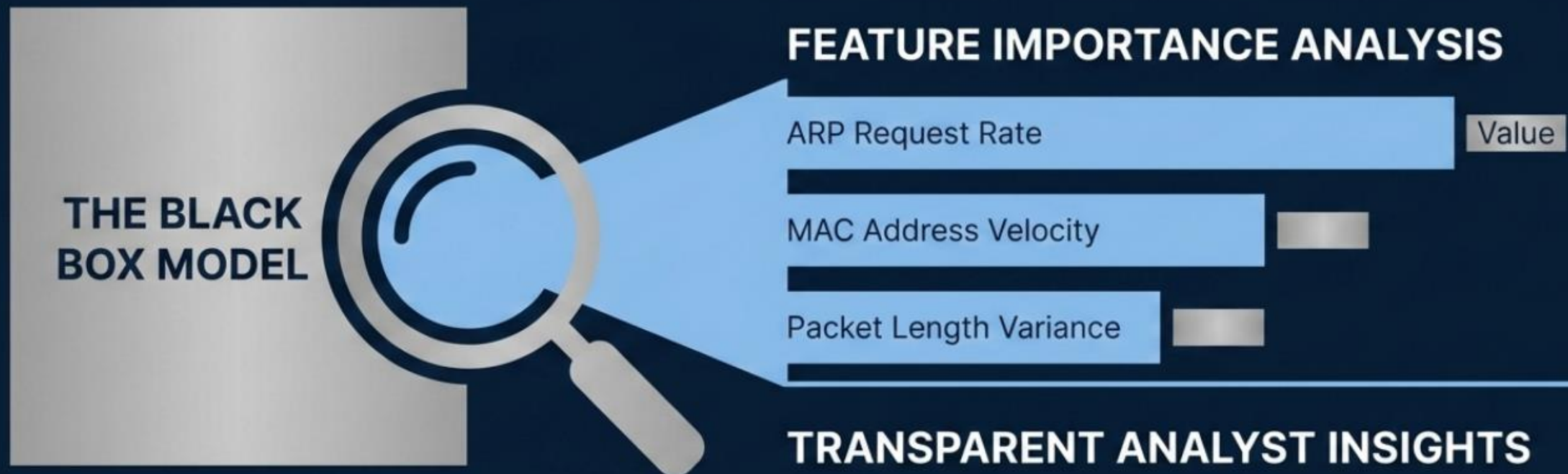
FEDPROX ALGORITHM

STABLE CONVERGENCE



Introduces a proximal term to balance unequal local training.

EXPLAINABLE AI (THE SHAP LENS)



METHODOLOGY (DATASETS)

Training & Simulation



CL2-IDS

Simulates Layer 2 behaviors across multiple federated nodes.

Real-World Validation



BCAST IDS

Introduces environmental noise to validate robustness.

Benchmarking



CICIDS-2017

Provides performance baselines against conventional legacy systems.

EVALUATION METRICS & HARDWARE VIABILITY



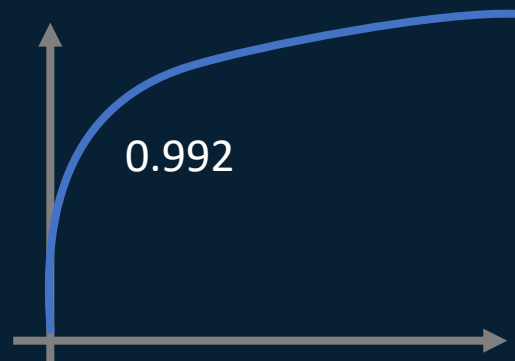
Accuracy



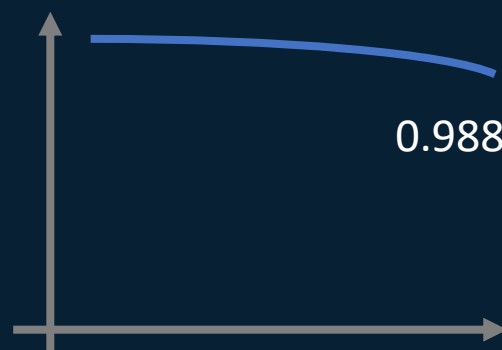
Precision



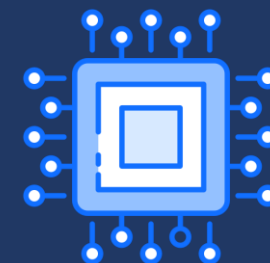
F1-Score



ROC-AUC Curve

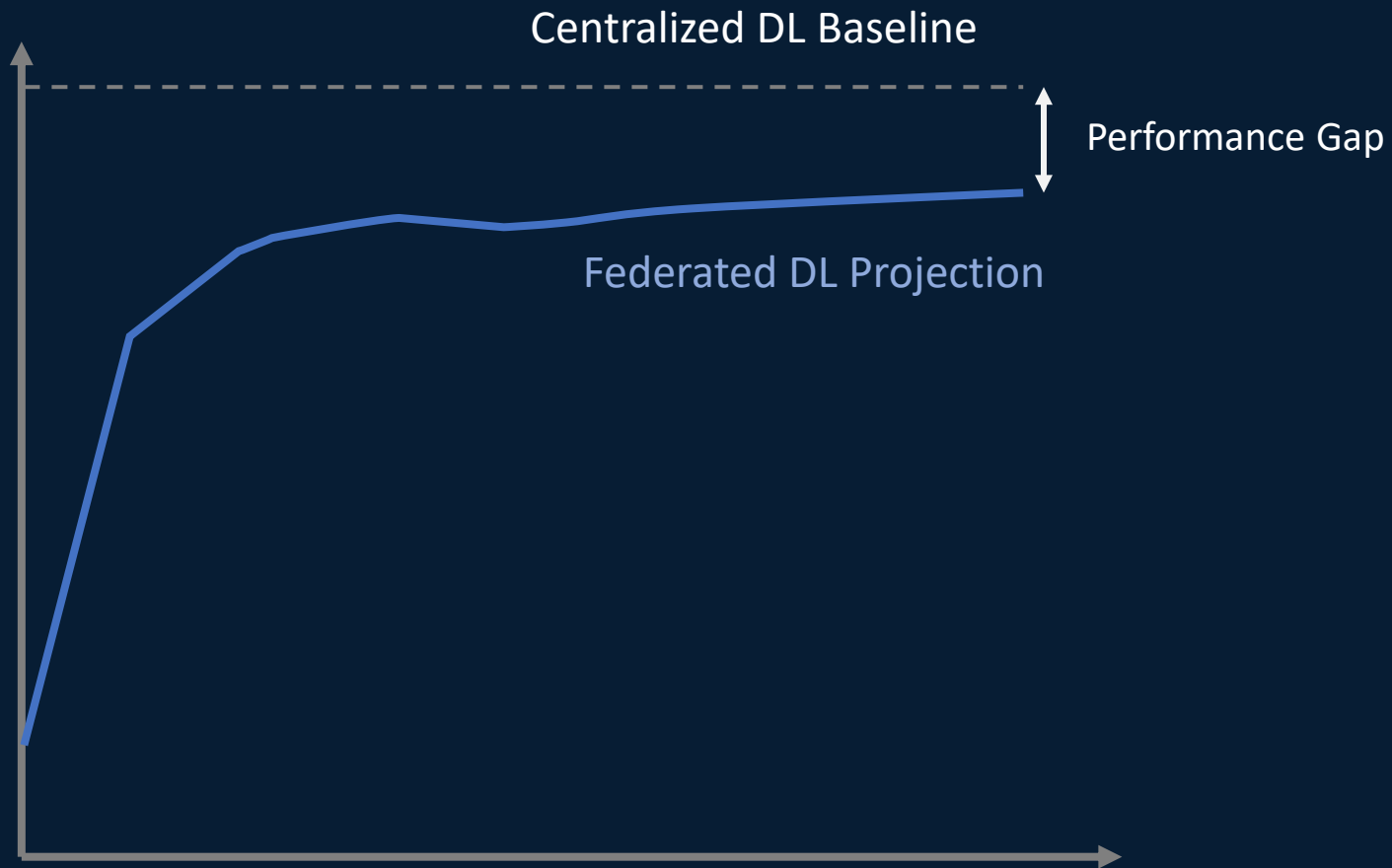


PR-AUC Curve



Hardware Utilization CPU &
Memory Overhead Monitoring

EXPECTED RESULTS & FEASIBILITY



**100% Data Privacy
Retained**

**Robust in Non-IID
Environments**

Edge Hardware Ready